

**STATE OF CALIFORNIA
CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION
895 Aerovista, Suite 101
San Luis Obispo, CA 9301**

**WASTE DISCHARGE/MASTER RECLAMATION REQUIREMENTS
ORDER NO. R3-2003-081**

**FOR
CITY OF SAN LUIS OBISPO
PRODUCER AND USER OF RECYCLED WATER
SAN LUIS OBISPO COUNTY**

The California Regional Water Quality Control Board, Central Coast Region (hereafter Board), finds:

SITE OWNER AND LOCATION

1. On November 16, 2001, David Pierce, Water Resources Manager of the City of San Luis Obispo (City), requested the Regional Board draft master reclamation requirements for the City of San Luis Obispo water reclamation plant. This Order also updates requirements in accordance with California Water Code Section 13523.1, which specifies requirements for supplying recycled water for inclusion in Master Reclamation Permits to supply recycled water.
2. The City of San Luis Obispo (Producer/User) (P.O. Box 1990, San Luis Obispo, CA 93102-1990) owns and operates water reclamation facilities on property owned by the Discharger in San Luis Obispo County (T31S, R12E, Section 10, MD B&M) as shown on Attachment "A".
3. Treatment and reclamation facilities include wet-weather flow equalization, screening, grinding, aerated grit removal, primary settling, biofiltration, secondary settling, nitrification using activated sludge, final settling, cooling using evaporative cooling towers, dual media filtration, and chlorination/dechlorination. Solids are thickened in a dissolved air floatation thickener, stabilized in anaerobic digesters and dewatered either by belt presses or drying beds. The treatment plant's design capacity (Average Dry Weather Flow) is 5.2 million gallons per day (MGD). Of this amount, the City is required by State Board Order 2000-07 DWR to maintain at least 1.1 mgd to San Luis Obispo Creek at all times. This currently leaves 2.9 MGD available for reclamation. Future City growth could increase wastewater flow to the plant to 5.2 mgd, allowing as much as 4.1 mgd to be reclaimed.
4. Currently identified reuse sites total 17 initial sites, two groundwater exchange/agricultural irrigation sites, and 4 potential future sites. Recycled water will be distributed through 8 miles of transmission/distribution lines, throughout the service area. These sites are identified as Attachments A and B. The City also provides/uses recycled water at construction sites throughout its service area for dust control and soil compaction.
5. **Surface water:** Surface drainage from the reuse sites is primarily via storm drains to the San Luis Obispo Creek watershed. This Order prohibits discharge to surface waters.

SITE/FACILITY DESCRIPTION

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BASIN PLAN

6. The Water Quality Control Plan, Central Coastal Basin, (Basin Plan) was adopted by the

Board on November 17, 1989 and approved by the State Board on August 16, 1990. The Board approved amendments to the Basin Plan on February 11, 1994 and September 8, 1994. The Basin Plan incorporates statewide plans and policies by reference and contains a strategy for protecting beneficial uses of State waters.

7. **Beneficial Uses:** Present and anticipated beneficial uses of ground water in the vicinity of the discharge include:

- a) Domestic,
- b) Municipal,
- c) Agricultural, and
- d) Industrial Supply.

8. **Ground Water Objectives:** Median water quality objectives specified in the Basin Plan for the San Luis Obispo Creek Sub-basin include:

Total Dissolved Solids	900 mg/l
Sodium	50 mg/l
Chloride	200 mg/l
Nitrate (as N)	5 mg/l
Sulfate	100 mg/l
Boron	0.2 mg/l

9. **Recycled Water:** State Department of Health Services' criteria for production and use of recycled water is in Title 22, Division 4, Chapter 3, of the California Code of Regulations. The Regional Board has consulted with the State and County Health Departments regarding regulation of this discharge

ENVIRONMENTAL SUMMARY

10. The Producer/User certified a final Environmental Impact for the Reclamation Project in March 1997, in accordance with the California Environmental Quality Act (Public Resources Code, Section 21000, et seq.) and the California Code of Regulations.

EXISTING ORDERS

11. The San Luis Obispo Water Reclamation plant is currently regulated by Waste Discharge

Requirements, Order No. R3-2002-0043 (NPDES No. CA0049224) which authorizes discharge to San Luis Obispo Creek.

12. Discharge of recycled water is a privilege, not a right, and authorization to discharge is conditional upon the discharge complying with provisions of Division 7 of the California Water Code and any more stringent effluent limitations necessary to implement water quality control plans, to protect beneficial uses, and to prevent nuisance. Compliance with this Order should assure this and mitigate any potential adverse changes in water quality due to the discharge.
13. On August 1, 2003 the Board notified the City and interested agencies and persons of its intent to issue water reclamation requirements for the discharge and has provided them with a copy of the proposed order and an opportunity to submit written views and comments.
14. After considering all comments pertaining to this use of recycled water during a public hearing on October 24, 2003, this Order was found consistent with the above findings.

IT IS HEREBY ORDERED, pursuant to authority in California Water Code Section 13523 et seq., the City of San Luis Obispo, its agents, successors, and assigns, may use recycled water as described above providing compliance is maintained with the following:

(Note: other prohibitions and conditions, definitions, and the method of determining compliance are contained in the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements" dated January, 1984)

WATER RECYCLING REQUIREMENTS

Use of Recycled Water shall be in conformance with recycled water criteria established in Title 22, Division 4, Chapter 3, of the California Code of Regulations.

A. PROHIBITIONS

1. Discharge of recycled water to areas other than approved reclamation areas (see Standard Provision A.6.) is prohibited.
2. Discharge of any recycled water including overflow, bypass, and overspray from transport, landscape irrigation reuse sites, agriculture, or construction sites to adjacent drainage ways or adjacent properties is prohibited. No runoff from any reclamation sites shall be permitted.
3. Bypass of the treatment facility and discharge of untreated or partially treated wastes directly to the reclamation areas is prohibited. Discharge of recycled water not in compliance with California Code of Regulations, Title 22, Division 4, Chapter 3, Article 4, Section 60313 (oxidized and adequately disinfected) is prohibited.

B. RECYCLED WATER SPECIFICATIONS

1. Effluent from the Water Reclamation facility shall meet all effluent limitations listed in Waste Discharge Requirements, Order No. R3-2002-0043 (NPDES No. CA0049224) before being further treated for reuse.
2. Daily discharge of recycled water shall not exceed 4.0 mgd.
3. Use of recycled water shall be accomplished at a time and in a manner that minimizes ponding and the possibility of public contact with sprayed materials.
4. Recycled water shall be applied when the grounds have maximum opportunity to dry before use by the public unless provisions are made to exclude the public from areas during irrigating with recycled water and while the areas are drying.
5. The median concentration of total coliform bacteria in the disinfected effluent shall not

exceed a Most Probable Number (MPN) of 2.2 per 100 milliliters, utilizing the bacteriological results of the last seven days for which analyses have been completed, and the MPN shall not exceed 23 per 100 mL in more than one sample in any 30-day period. No sample shall exceed an MPN of 240 total coliform bacteria per 100 mL.

6. The filtered wastewater shall be disinfected by a chlorine disinfection process that provides a CT (chlorine concentration times modal contact time) value of not less than 450 milligram-minutes per liter at all times with a modal contact time of at least 90 minutes, based on peak dry weather flow.
7. Drinking water fountains shall be protected from direct or windblown recycled water spray.
8. Recycled water shall be applied at a rate and volume not to exceed vegetative demand and soil moisture conditions needed to leach excess salts to below the root zone. Special precautions must be taken to prevent clogging of spray nozzles, to prevent overwatering and to minimize the production of runoff. Pipelines shall be maintained so as to prevent leaks.
9. Recycled water shall not be used for irrigation during periods of extended rainfall and/or runoff.
10. Signs shall be provided at reuse sites to inform the public that recycled water is being stored or used. For golf courses, notification shall also be placed on scorecards.
11. Personnel involved in transporting or using recycled water shall have undergone adequate training in recycled water use, as provided by the Producer/Use.
12. Recycled water piping (excluding existing landscape piping), controllers, valves, etc.,

shall be marked to differentiate recycled water facilities from potable water facilities.

13. Recycled water valves, outlets, quick couplers, and sprinklers shall be of a type, or secured in a manner, that permits operation only by authorized personnel.
14. Use or installation of hose bibs on the recycled water system shall not be permitted.
15. Proper backflow and cross-connection protection for domestic water services and irrigation wells shall be provided.
16. Recycled water shall not be irrigated or impounded within 100 feet of any well used for domestic supply.
17. Recycled water systems shall be inspected quarterly to assure proper operation, absence of leaks, and absence of illegal connections.

C. USE FOR CONSTRUCTION PURPOSES

The Producer/User shall ensure that recycled water is used at construction sites for dust control or soil compaction in accordance with the following requirements:

1. Producer/User shall instruct, orally and in writing, each recycled water tank truck driver as to the requirements of this Order and the potential health hazards involved with the reuse of wastewater.
2. The Producer/User shall ensure that tank trucks and other equipment which contain or contact recycled water are clearly identified with warning signs which state, for example, "RECYCLED WATER - DO NOT DRINK".
3. The User shall ensure that tank trucks are thoroughly cleaned of septage or other contaminants before and after use.

4. Recycled water shall be confined to the authorized use area.
5. Recycled water shall be applied to reuse sites so as to ensure that no ponding or runoff occurs.
6. Recycled water shall be applied to reuse sites so as to minimize aerosol formation during spraying.
7. Recycled water shall be applied to reuse sites so as to prevent public or employee contact with the water.
8. The Producer/User shall ensure that recycled water cannot be introduced into any permanent piping system, and that no connection can be made between the tank truck and any part of a domestic water system.
9. The Producer/User shall ensure that reuse sites are posted with signs informing the site personnel that recycled water is in use.

D. PROVISIONS

1. Producer/User shall comply with "Monitoring and Reporting Program No. R3-2003-081", as specified by the Executive Officer.
2. Producer/User shall comply with all items of the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements" dated January, 1985, except Item Nos. A.11., A.14., A.15., B.4, C.4., C.9, D.1. and D.2.
3. At least 30 days prior to reuse, the Producer/User shall submit plans and specifications to this Board and the State Department of Health Services, which show how compliance with prohibitions and recycled water specifications in this Order will be ensured.

4. Prior to water reclamation at landscape irrigation reuse sites, the Producer/User shall:

- a. Obtain approval of the plans and specifications required by Provision D.3 from the Executive Officer after consultation with the State Department of Health Services. (The Regional Board shall provide comments on, or approval of the producer/user proposal within the 30-day review period, or approval shall be deemed granted.)
- b. Enter into a contractual agreement (User Agreement) with all recycled water users to ensure all parties understand the proper water reuse practices needed to comply with this Order. A User Manual shall be provided to each site supervisor.

5. Pursuant to California Code of Regulations Title 23, Chapter 3, Subchapter 9, the Producer/User shall submit a written report to the Executive Officer no later than April 24, 2008 addressing:

- a. Whether there will be changes in the continuity, character, location, or volume of the discharge; and,
- b. Whether, in their opinion, there is any portion of the Order that is incorrect, obsolete, or otherwise in need of revision.

I, ROGER W. BRIGGS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Coast Region, on October 24, 2003.

/S/
Executive Officer